

Ohio Department of Health – Bureau of Vital Statistics
Guide to Mortality Data in the Ohio Public Health Information Warehouse

Introduction:

The Ohio Public Health Information Warehouse application stores large volumes of vital data to support ongoing activities such as surveillance, assessments, grant writing, and evaluations. The application improves the efficiency and quality of Ohio Department of Health (ODH) dissemination practices creates timely decision support and analytic tools (i.e. geo-spatial, etc.). This document provides a background for data users and describes the documentation available to support data use.

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Data Source:

ODH receives Certificates of Death after filing at the local health departments for all the deaths occurring in Ohio. Funeral homes enter demographic data directly into Ohio's Electronic Death Registration System (EDRS). Physicians and Coroners report decedent medical information. The EDRS application collects both the demographic and medical information as recommended by the National Center for Health Statistics (NCHS) in the 2005 death certificate revision. Electronic specifications can be viewed at: http://www.cdc.gov/nchs/nvss/vital_certificate_revisions.htm

ODH compiles these records then sends key data to NCHS as weekly extracts. NCHS incorporates the Ohio files into the national Vital Statistics database. NCHS also returns classification data to Ohio for the assignment of death cause codes. Warehouse download files are subsequently created from the Ohio database.

The Warehouse death dataset contains virtually all variables in the Ohio reporting database. The death dataset contains events which occurred in Ohio from 2007 to the present (1.5+ million records). It also contains records from other jurisdictions which shared data for an Ohio resident who died outside of Ohio. (approximately 2% of Ohio deaths)

Mortality data is updated nightly. Data downloaded today may include different records and values tomorrow. Data are in flux until an entire calendar year is locked down (closed). Once a year is closed, further updates won't occur and the data will remain stable.

At some point in the following year, a calendar year of data will be considered complete. At that point we will stop the flow of data into the Warehouse so that the file will remain unchanged. Until that time, data are considered preliminary. Preliminary data can inform important policy decisions, potentially leading to improved health outcomes. All preliminary data in the Warehouse will have a footnote stating that the data is preliminary and subject to change. Incomplete variables will populate with information as it is received. All calculated rates will be suppressed for partial years.

Charts:

Web users have the ability to customize a chart or graph using Ohio mortality data.

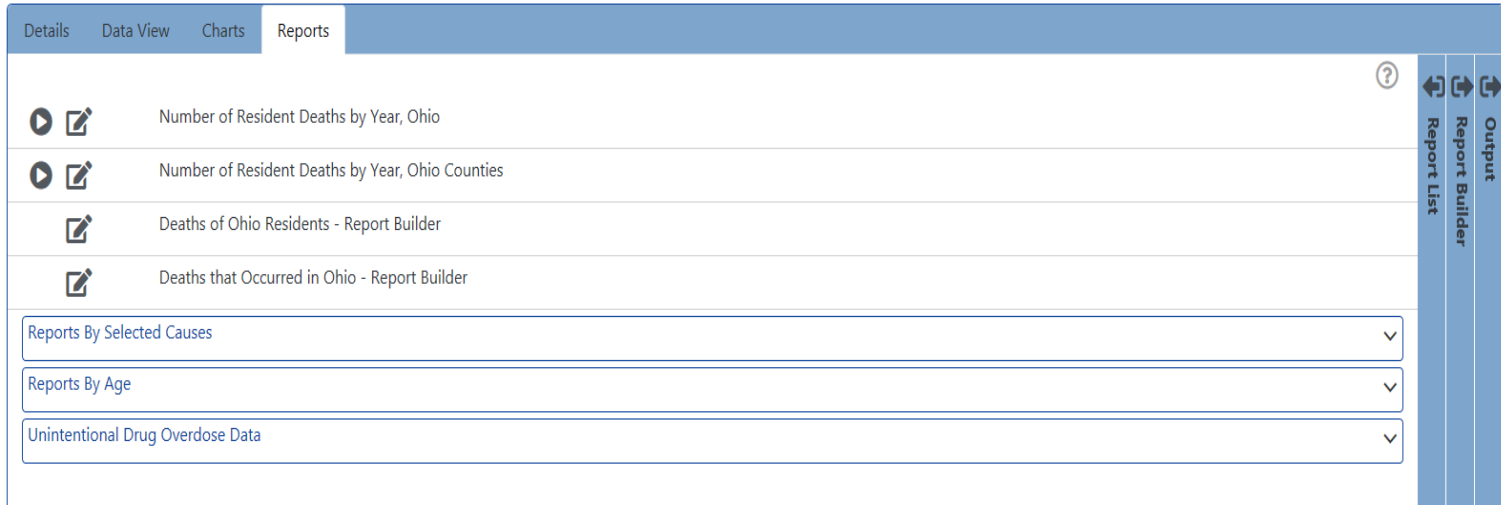


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Reports:

Web users also have the ability to Run or Customize numerous reports.



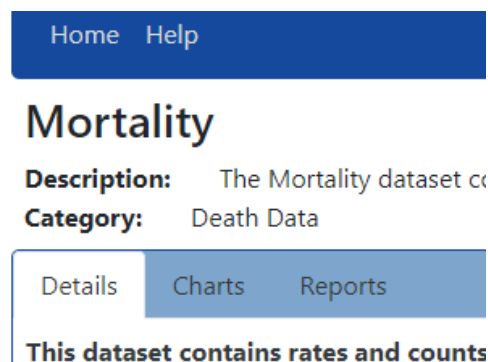
Additional Information for Data Users:

COVID-19 Pandemic

The CDC recommended 'Coronavirus Disease 2019' or 'COVID-19' should be reported on the death certificate for all decedents where the disease caused or is assumed to have caused or contributed to death. Detailed reporting guidance is available at <https://www.cdc.gov/nchs/covid19/coding-and-reporting.htm>

The counts of COVID-19 deaths obtained in the ODH Warehouse are based on cause of death statements contained on death certificates and will not exactly line up with other sources of COVID-19 mortality counts, such as those based on COVID-19 testing.

The Warehouse enables users to tabulate Ohio resident and occurrence deaths for custom requests. From the Mortality module, select the 'Reports' tab.



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For resident tabulation, choose: Deaths of Ohio Residents – Report Builder

For occurrence tabulation, choose: Deaths that Occurred in Ohio – Report Builder

Category: Death Data

Details	Charts	Reports
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		Number of Resident Deaths by Year, Ohio
		Number of Resident Deaths by Year, Ohio Counties
		Deaths of Ohio Residents - Report Builder
		Deaths that Occurred in Ohio - Report Builder

Reports By Selected Causes

Select your report criteria with Filter, Row, Column, and Measure choices:

Details	Charts	Reports
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 **Report List**

Deaths of Ohio Residents - Report Builder

Filter Variables

Select Row and Column Variables

Select Measures

Report Options

Show ReportExport Data

Filtering for the report narrows the table request to your needs. The best way to select for COVID-19 deaths is to filter the report using: Specific Causes of Death (358)

COVID-19 is coded to International Classification of Diseases, Revision 10 code U071 which is contained in the cause choice clipped above. Virtually all the deaths in the category will be COVID-19 deaths.

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Further selecting row and column variables will result in your table.

Deaths of Ohio Residents - Report Builder

Filter Variables

Category

Variable

Values

Causes of death

Specific Causes of Death (358)

Search...

+

Select Row and Column Variables

Select Measures

Report Options

Show Report Export Data

☒ Select All

☐ Acute bronchitis and bronchiolitis (J20-J21)

☒ Unspecified acute lower respiratory infection(J22,U049,U071)

☐ Other diseases of upper respiratory tract (J30-J39)

☐ Bronchitis, not specified as acute or chronic(J40)

☐ Chronic bronchitis (J41-J42)

The report clipped below is the report after filtering for year=2020, Specific Causes of Death (358) = Unspecified acute lower respiratory infection (J22, U049, U071), and requesting Age of decedent in rows and month of death in columns:

Ohio Resident Deaths

List of data filters applied in this report:

by Month, Age 11 Categories

for Specific Causes of Death (358) = (Unspecified acute lower respiratory infection(J22,U049,U071)) , Year = (2020 **) , Res State = (OH)

Month of Death

Age 11 Categories

Month	January	February	March	April	May	June
Age 11 Categories	Death Count	Death Count	Death Count	Death Count	Death Count	Death Count
< 1	0	0	0	0	0	0
1-4	0	0	0	0	0	0
5-14	0	0	0	0	0	0
15-24	0	0	0	0	0	0
25-34	0	0	0	0	0	0
35-44	0	0	1	1	0	0
45-54	0	0	3	4	0	0
55-64	0	0	11	7	0	0
65-74	0	0	9	27	0	0
75-84	0	0	15	30	0	0
85+	0	0	17	37	0	0
Unk	0	0	0	0	0	0
Total	0	0	56	106	0	0

Ages grouped into 10 year intervals

Ohio Resident deaths include individuals that resided in Ohio at the time of death regardless of where the death occurred.

Excludes 0 Deaths and 0 Population records with unknown/missing data.

Years with ** are considered partial and may be incomplete.

* - Indicates that the cell value has been blinded to protect confidentiality.

N/A - Indicates rates have been suppressed for counts < 10 or where population counts are not available, rates based on counts < 20 are considered unreliable.

Report created in the Ohio Public Health Data Warehouse on 5/5/2020 9:57 AM with data from 5/4/2020

The Ohio Department of Health specifically disclaims responsibility for any analysis, interpretations or conclusions.

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Special note on ICD 10 Cause List codes

NCHS has recommended using the standard cause codes to present the underlying cause of death for tabulation and dissemination of mortality data. NCHS periodically updates these cause code lists based on the WHO Mortality tabulation lists. ICD-10 is the latest cause code list that is used presently.

The cause of death information is submitted on death certificates and supplemental death certificates in terms of literals. ODH sends the death data records to NCHS from the EDRS system. After processing these records, NCHS sends back ICD-10 codes and other cause of death information. Cause of death codes were determined automatically by processing ACME code.

For the Warehouse, we classified all deaths into groupings based on cause of death, using the NCHS 113 cause list, 130 Infant cause list, 358 cause list, and 39 cause list. These lists present a standard set of causes to display. In addition, the lists identify these groupings that are rankable as a leading cause of death. The 113 Selected Causes of Death and 130 Selected Causes of Death lists are further grouped into two condensed lists 113 Leading Cause of Death and 130 Leading cause of Death. These codes help in identifying the leading cause of death.

- List of 358 Selected Causes of Death : To comply with WHO Tabulation Regulations
- List of 113 Selected Causes of Death: For data tabulation and analysis of general mortality and ranking leading causes of death
- List of 130 Selected Causes of Death: For data tabulation and analysis of infant mortality and ranking leading causes of infant death
- List of 39 Selected Causes of Death: For tabulations by smaller geographic areas
- List of 113 Leading causes of Death: For identifying leading cause of death
- List of 130 Leading causes of Death: For identifying leading cause of death in infant deaths

The cause of death coding system currently used in the United States is described in the International Classification of Diseases, Tenth Revision <https://www.cdc.gov/nchs/data/dvs/Part9InstructionManual2011.pdf>

Special note about race and ethnicity

Race and ethnicity are reported on the death certificate in terms of literals and check boxes. The data are entered into EDRS as a series of Yes/No checkboxes and open-ended literal blocks. There are 14 specific race checkboxes and 4 specific Hispanic ethnicity checkboxes. Additionally, one literal ethnicity checkbox is provided and several literal race checkboxes are provided (e.g. American Indian tribe). These entries yield a multi-race view of the decedent. The collected checkbox and literal race data are submitted weekly (along with the other death data) to NCHS. NCHS then codes, edits, and returns results to ODH. The returned multi race files contain 3-digit codes analogous to those used in the US Census. There may be up to 8 codes per person in this returned file. NCHS code lists for race are accessible in PDF at this website: <https://www.cdc.gov/nchs/data/dvs/racecodelist.pdf>

NCHS code lists for ethnicity are accessible in PDF at this website:
https://www.cdc.gov/nchs/data/dvs/Race_Ethnicity_CodeSet.pdf

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The NCHS race recoding system also bridges multiple-race data into the single race format. This allows for analyses of vital statistics data on a single race. The bridging methodology can be downloaded at:

https://www.cdc.gov/nchs/nvss/bridged_race.htm

The bridged race values are used to create several more race values for each death. We used the decedent's bridged race value from NCHS to create the single race tabulation variable Race_Category. Race_Category is used in Warehouse race tables and used to group decedent's race into the following categories: White, Black, Native American, Asian, Pacific Islander/Hawaiian, and Other/Unknown. While we await return from NCHS of the matching race file, the Warehouse assigns a tentative Race_Category value to each death record based on check box race data. We bridge the multi-race checkbox data to single race by assigning the least populous race to the mothers that report multiple races.

We offer very granular views of race and ethnicity. Users are reminded to be careful of publishing tables crossed by race/ethnicity. Some races are very rare in many parts of Ohio. If users release statistics by race there is a risk of disclosure of personal information about the death in the rare race group.

Special note on Geocoding Fields

Geocoding is the process of enriching a description of a location; most typically a postal address with geographic coordinates from spatial reference data. The Ohio Department of Administrative Services geocoding routine is invoked on the death records to save the geocoding fields into the system. Our process in the past used to geocode only location of residence address fields now we extended the process to gather geocoding data for location of death, location of Injury and location of residence for out of state records.

Special note on residence

Decedent residence data is reported by the funeral home on the death certificate. Decedent's residence street address, city, zip code, and county of residence fields are saved to EDRS. The Ohio Department of Administrative Services geocoding routine is invoked when the death record is saved. Our process in the past and continuing today is to use the results of the geocoding service to populate the location fields into the warehouse data.

Citation:

Please use the following citation in any publication or release which uses or references data from the Warehouse: "These data were provided by the Ohio Department of Health. The Department specifically disclaims responsibility for any analyses, interpretations or conclusions".